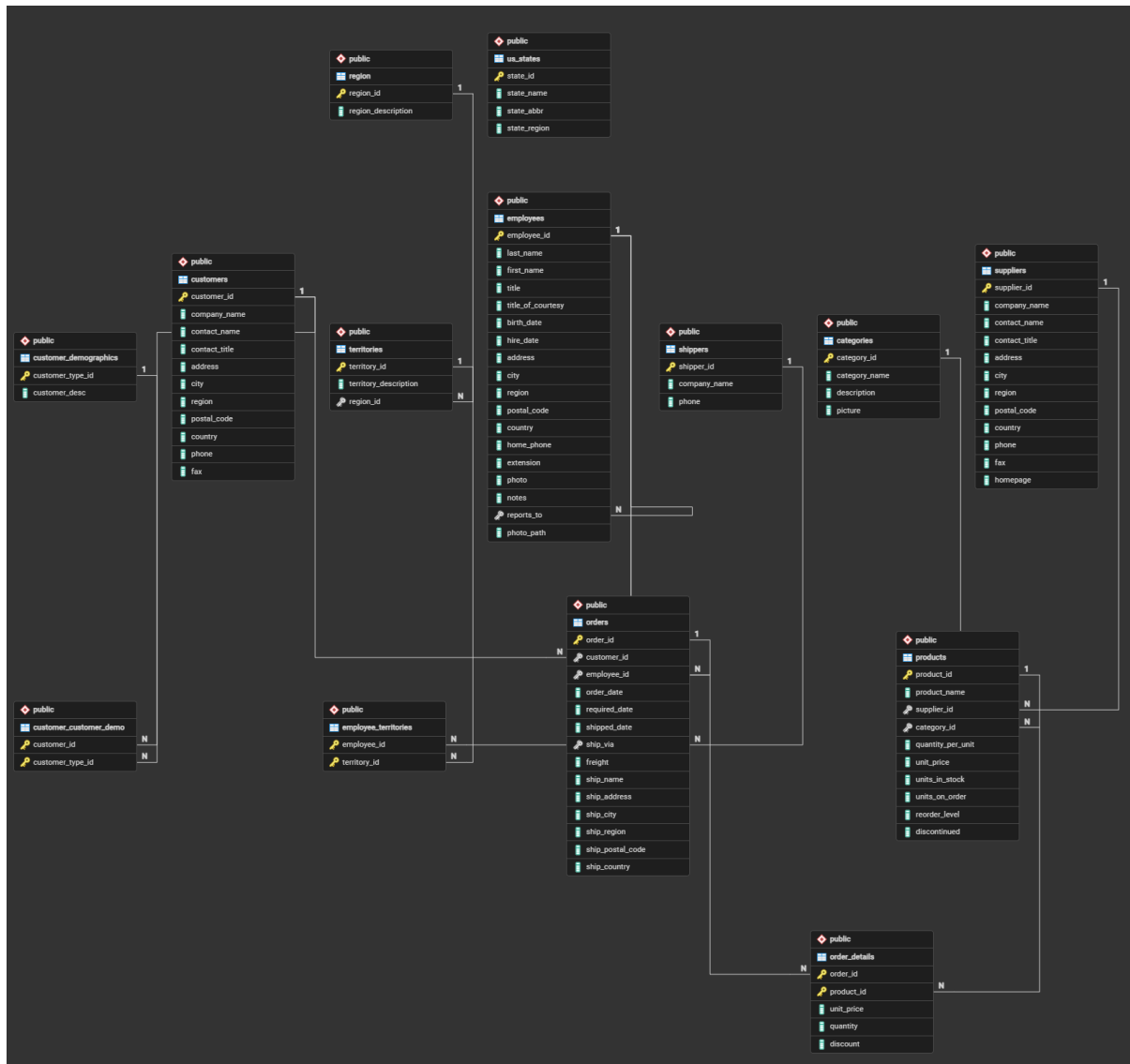


In-term Test Q4 2025-2026



ER Model

Question 1 (10 marks): Database Setup (10 marks)

Using PostgreSQL and pgAdmin:

(a) Create a database named yourfullname. Then move all Northwind tables into this schema. (5 marks)

(b) Create a new table called students inside schema exam with the following columns:

Studentid: 5-digit number, Primary Key

Fullname: Required

Email: Must be unique

Insert your own information into the table. (5 marks).

Output:

Results 1: Verify result from load and create database

The screenshot shows the pgAdmin 4 interface with the following content:

Query Editor:

```

25 Insert your own information into the table. (5 marks)
26
27 Then you can check your database before continuing.
28
29 /*
30 SELECT table_name, column_name, data_type
31 FROM information_schema.columns
32 WHERE table_schema = 'public' |
33 ORDER BY table_name, ordinal_position;
34
35 --(b) Create table students
36 CREATE TABLE exam.students (
37     studentid CHAR(5) PRIMARY KEY,
38     fullname VARCHAR(100) NOT NULL,
39     email VARCHAR(100) UNIQUE NOT NULL;
40
41 --(c) Insert your own record
42 INSERT INTO exam.students (studentid, fullname, email)
43 VALUES ('00044', 'VU DONG DUONG', 'duong.vu.bbs20@fu.edu.vn');
44
45 -- (d) Verify the result
46 SELECT * FROM exam.students;

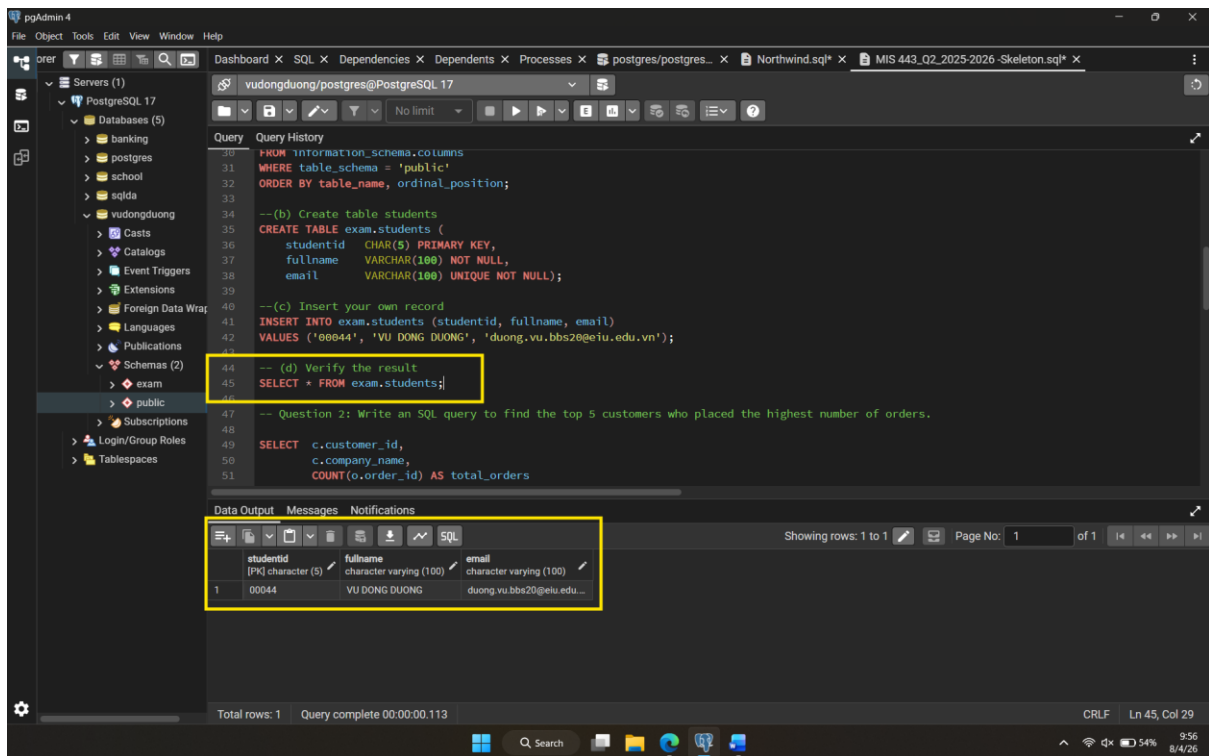
```

Data Output:

table_name	column_name	data_type
categories	category_id	smallint
categories	category_name	character varying
categories	description	text
categories	picture	bytea
customer_customer_de...	customer_id	character varying
customer_customer_de...	customer_type_id	character varying

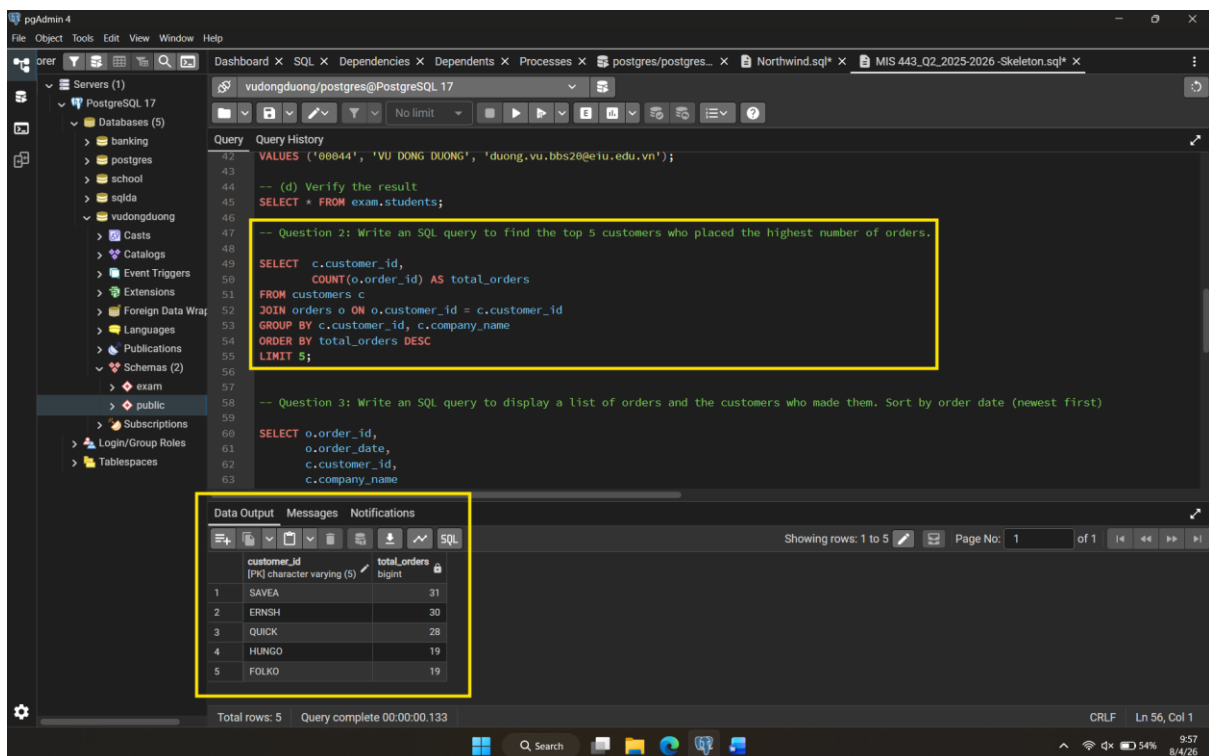
Total rows: 92 Query complete 00:00:00.098

Results 2: Verify result from table student



Question 2 (10 marks): Write an SQL query to find the top 5 customers who placed the highest number of orders.

Output:



Question 3 (20 marks): Write an SQL query to display a list of orders and the customers who made them. Sort by order date (newest first).

Output:

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'PostgreSQL 17', including databases like 'banking', 'postgres', 'school', 'sqlda', and 'vudongduong'. The main pane shows the 'Query History' tab with a query for 'vudongduong/postgres@PostgreSQL 17'. The query is highlighted with a yellow box:

```
-- Question 3: Write an SQL query to display a list of orders and the customers who made them. Sort by order date (newest first)

SELECT o.order_id,
       o.order_date,
       c.company_name
FROM orders o
JOIN customers c ON c.customer_id = o.customer_id
ORDER BY o.order_date DESC;
```

Below the query, the 'Data Output' tab is also highlighted with a yellow box, showing the results of the query. The results are displayed in a table with columns: 'order_id', 'order_date', and 'company_name'. The table shows 6 rows of data, with the first row being order_id 11074, order_date 1998-05-06, and company_name 'Simons bistro'.

order_id	order_date	company_name
11074	1998-05-06	Simons bistro
11077	1998-05-06	Rattlesnake Canyon Grocery
11076	1998-05-06	Bon app'
11075	1998-05-06	Richter Supermarkt
11071	1998-05-05	LILA-Supermercado
11073	1998-05-05	Pericles Comidas clásicas

The bottom status bar indicates 'Total rows: 830' and 'Query complete 00:00:00.111'.

Question 4 (20 marks): Northwind management wants to identify large product movements to better plan inventory and logistics. Write an SQL query to display orders where a product was purchased in large quantity (more than 99 units in a single order).

Output:

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'public' schema. The main query editor contains the following SQL code:

```

-- Question 4: Northwind management wants to identify large product movements to better plan inventory and logistics.
-- Write an SQL query to display orders where a product was purchased in large quantity (more than 99 units in a single order).

SELECT o.order_id,
       o.order_date,
       p.product_name,
       od.quantity
FROM order_details od
JOIN orders o ON o.order_id = od.order_id
JOIN customers c ON c.customer_id = o.customer_id
JOIN products p ON p.product_id = od.product_id
WHERE od.quantity > 99
ORDER BY od.quantity DESC;

```

The 'Data Output' tab shows the results of the query, displaying 6 rows of data:

order_id	order_date	product_name	quantity
11072	1998-05-05	Wimmers gute Semmelknödel	130
10764	1997-12-03	Chartreuse verte	130
10451	1997-02-19	Pâté chinois	120
10515	1997-04-23	Schoggi Schokolade	120
10711	1997-10-21	Perth Pasties	120
10398	1996-12-30	Pâté chinois	120

The status bar at the bottom indicates: 'Total rows: 23', 'Query complete 00:00:00.207', and 'Successfully run. Total query runtime: 207 msec. 23 rows affected.'

Question 5 (20 marks): Northwind management wants to evaluate the delivery performance of each shipping partner. Write an SQL query to calculate the average delivery time (in days) for each shipper. Delivery time = shipped_date – order_date.

Output:

The screenshot shows the pgAdmin 4 interface with the following content:

Query History:

```

75 FROM order_details od
76 JOIN orders o ON o.order_id = od.order_id
77 JOIN customers c ON c.customer_id = o.customer_id
78 JOIN products p ON p.product_id = od.product_id
79 WHERE od.quantity > 99
80 ORDER BY od.quantity DESC;
81
82 -- Question 5: Northwind management wants to evaluate the delivery performance of each shipping partner.
83 -- Write an SQL query to calculate the average delivery time (in days) for each shipper. Delivery time = shipped_date - order_date
84
85 SELECT s.company_name,
86        AVG(o.shipped_date - o.order_date) AS avg_delivery_days
87 FROM orders o
88 JOIN shippers s ON s.shipper_id = o.ship_via
89 WHERE o.shipped_date IS NOT NULL
90 GROUP BY s.shipper_id, s.company_name
91 ORDER BY avg_delivery_days DESC;
92
93
94 -- Question 6: Northwind wants to identify the most active customers (customers who place orders most frequently) to target retention campaigns
95 SELECT c.customer_id,
96

```

Data Output:

	company_name character varying (40)	avg_delivery_days numeric
1	United Package	9.2349206349206349
2	Speedy Express	8.5714285714285714
3	Federal Shipping	7.4738955823293173

Total rows: 3 Query complete 00:00:00.123

Question 6 (20 marks): Northwind wants to identify the most active customers (customers who place orders most frequently) to target retention campaigns. Write an SQL query to rank customers based on their total number of orders (highest = rank 1). Customers with the same number of orders must have the same rank.

Output:

The screenshot shows the pgAdmin 4 interface with two SQL queries executed. The first query calculates the average delivery time for each shipping partner. The second query ranks customers by the number of orders they placed.

Query 1: Average delivery time by shipping partner

```
-- Question 5: Northwind management wants to evaluate the delivery performance of each shipping partner.
-- Write an SQL query to calculate the average delivery time (in days) for each shipper. Delivery time = shipped_date - order_date

SELECT s.company_name,
       AVG(o.shipped_date - o.order_date) AS avg_delivery_days
FROM orders o
JOIN shippers s ON s.shipper_id = o.ship_via
WHERE o.shipped_date IS NOT NULL
GROUP BY s.shipper_id, s.company_name
ORDER BY avg_delivery_days DESC;
```

Query 2: Rank customers by total orders

```
-- Question 6: Northwind wants to identify the most active customers (customers who place orders most frequently)
-- to target retention campaigns. Write an SQL query to rank customers based on their total number of orders (highest = rank 1).
-- Customers with the same number of orders must have the same rank.

SELECT c.customer_id,
       c.company_name,
       COUNT(o.order_id) AS total_orders,
       RANK() OVER (ORDER BY COUNT(o.order_id) DESC) AS customer_rank
FROM customers c
JOIN orders o ON o.customer_id = c.customer_id
GROUP BY c.customer_id, c.company_name
ORDER BY customer_rank;
```

Data Output:

customer_id	company_name	total_orders	customer_rank
1	SAVEA	31	1
2	ERNSH	30	2
3	QUICK	28	3
4	HUNGO	19	4
5	FOLKO	19	4
6	BERGS	18	6

Total rows: 89 Query complete 00:00:00.175

----- End -----